

New Laws, New Theorems, New Physics

The constructive turn: the unified corpus held to the demand for substantial and conceptually novel advances. Four laws of bounded observation legislated from the corpus's scattered no-go theorems; six new theorems with their proved halves separated from their open cores; record phase transitions in monitored quantum circuits; an obstruction index measurable in hardware; and five instrument-grade capabilities, each with a protocol and a falsifier.

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1 · The Mandate: New Science, Not New Vocabulary

Volume I built the dictionary. Volume II adjudicated the theorems, corrected them, and isolated their open cores — and both volumes ended at the same place: a frontier, with every open link named and no new mathematics claimed. The mandate behind this volume refuses to stop there. It demands new physics, new theorems, new laws — substantial and conceptually novel advances in understanding or in technological capability — and it is explicit that a synthesis which merely renames its ingredients does not qualify. This volume is therefore the constructive turn: it takes the unified object the first two volumes exposed, and creates from it the strongest new science its honesty discipline permits.

We want new physics, new theorems, new laws. Substantial and conceptually novel advances in understanding and/or technological capability.

The discipline is fixed before anything is created, because the corpus itself enforces it. Every statement below carries exactly one of five provenance tags. **PROVED-CORPUS**: a result already in the manuscripts, cited by label and version, with nothing added. **PROVED-HERE**: a new result whose proof obligation is discharged in this volume, either directly or by applying a standard theorem to corpus objects — Birkhoff's theorem, the Cherny–Spohn criterion, the singular-value product law — with the application stated precisely enough to be checked. **DERIVED**: a new object computed from the corpus's exact formulas, arithmetic only. **CONJECTURE**: a new statement with named witnesses and no proof. **PREDICTION**: a falsifiable empirical claim with a named experiment and a verdict condition. The one-honest-open-link rule of Volume II carries over unchanged: no theorem below claims more than its weakest proved ingredient, and Table 2 separates every theorem's proved halves from its open core in print.

Novelty itself is graded, and the grades are declared rather than implied. A new statement is a theorem the corpus does not contain; a new proof route is a discharged hypothesis that a corpus theorem currently carries; a new object is an invariant the corpus never assembled; a new prediction is a number not yet measured anywhere; a new instrument is a protocol that did not exist. This volume delivers at least one of each grade — four laws (new statements), two discharged hypotheses (Theorems IV and VI each discharge or weaken one), one assembled invariant (the conserved obstruction), nine predictions, and five instruments. The measure of success is external: DeepSeek's verdict on the whole programme stands — until the unified framework proves something no ingredient could prove alone, the merger remains a hypothesis — so the volume ends with a ledger, not a peroration: nine falsifiable entries, each with its refuter named.

2 · The Four Laws of Bounded Observation

Thermodynamics is the precedent, and it is the right one. Before Carnot and Clausius, the science of engines was a pile of true facts — no engine exceeds Carnot efficiency, no process un-mixes heat spontaneously, no perpetual motion machine works — each proved or observed locally, none governed by a single statement. The laws of thermodynamics legislated that pile: four short statements from which the facts reorganized as instances, and from which new predictions fell out. The corpus is in exactly the pre-legislation position. It contains no right adjoint for environment tensoring, no universal evaluator, AAK floors that no finite machine beats, a capability problem complete for Σ^3 , sixty-six percent of metabolic cycles that do not revert, a contraction at 0.697, and a flat-width floor of $4/3$ — all proved, all scattered, all describing one object: the bounded observer and what it cannot do. This chapter legislates them.

THE CORPUS AS IT STANDS — PROVED, SCATTERED

no right adjoint no universal evaluator AAK floor σ_{k+1} 66% non-reversion
 Σ^3 -complete capability 0.697 contraction flat width $\geq 4/3$

All true, all proved, all scattered: no single statement governs them. The constructive turn legislates them — the way the pile of engine facts became the laws of thermodynamics.

THE FOUR LAWS OF BOUNDED OBSERVATION

00 Equilibrium blindness Cherny-Spohn · no-freeze degenerate case · ECOMOD eigenvalue
 at equilibrium the observer's obstruction is inert — no distortion produced, no arrow of time detectable, the bottleneck invisible from inside

▼ **DERIVED + PREDICTION** off equilibrium, obstruction becomes observable — and productive

01 Conservation of obstruction Theorem I · intercept · no-programming · Borsuk-Ulam
 ignorance relocates (repair), refines (descent), prices (rate-distortion) — never dies; leakage $\delta(e) = e^2 - 1$, multiplicative along cascades

▼ **THEOREM I** the conserved quantity makes the no-go theorems its instances

02 Distortion production Theorem III · slope 1.00 · $3/2$ fatigue · replica gap 0.0224
 nonzero obstruction $\Rightarrow$ irreversible distortion at Cesaro rate $\dot{D}$, floored by the AAK tail; off equilibrium, $\dot{D}$ is the observer's entropy production

▼ **THEOREM III · (d3) CONJECTURE** the arrow of time for bounded observers — five witnesses across three subcorpora

03 Unattainability WKL / PA degrees · PostBQP · AAK tail · Borsuk-Ulam
 zero obstruction at finite price is unreachable: the spectral tail prices the approach, compactness failure prices the certificate search — two walls, one phenomenon

FOUR LAWS each grounded in proved results on one side and a falsifiable prediction on the other — the refuters are named in Table 3

Figure 1 — The four laws of bounded observation: the corpus's scattered no-go theorems and bounds, legislated. Each law carries a proved face and a predictive face; the refuters are named in Table 3.

Law	Statement	Corpus grounding	Status and refuter
0 — Equilibrium blindness	At equilibrium a bounded observer's obstruction is inert: no distortion is produced, no arrow of time is detectable from inside; obstruction exists only off equilibrium.	Cherny–Spohn reversibility criterion (zero entropy production iff reversible); the degenerate zero-variance case of the no-freeze theorem; ECOMOD's constant leading eigenvalue (-0.0537 per year) along the drawdown branch; the uniform-continuity duality of the profinite limit.	DERIVED; refuted by any equilibrium observer that detects its own bottleneck (Prediction 7)
1 — Conservation of obstruction	Ignorance is conserved under exact transport: it can be relocated, refined, or priced, never destroyed; the leakage of a non-exact transport is measurable and multiplies along cascades.	Obstruction-datum transport and overhead spectrum transport (ECD v47); the intercept classification (no adjoint except trivial environment); the no-programming theorem; the adjunction defect $\delta(e) = e^2 - 1$ with multiplicative rigidity.	Theorem I; the conservation form and the cascade law are new; every ingredient proved
2 — Distortion production	A bounded observer with nonzero obstruction produces distortion irreversibly; its Cesaro rate is floored by the AAK tail and, off equilibrium, equals the observer's entropy production.	The exact fixed-budget rate–distortion formula with greedy achievability (Thm 6.6); the zero threshold $\text{rank}(H) \leq M$ (Prop 7.8); AAK floors (Thm 7.9); the replica interpolation identity; slope-1.00 non-reversion; the $3/2$ fatigue law.	Theorem III; parts (a)–(c) and (d1)–(d2) proved here; the general equality (d3) conjectured with witnesses
3 — Unattainability	Zero obstruction at finite price is unreachable: the approach is priced by the spectral tail and the certificate search by compactness failure — two faces of one compactness phenomenon.	AAK/Nehari floors; the Borsuk–Ulam compression floor $2/e^2$; the WKL and PA-degree capability spectra; Σ^3 -completeness of recursive capability; PostBQP-completeness of trajectory decisions; the memory floor $\liminf (\dim M(e)^2 - 1) \geq e^2 - 1$.	HYPOTHESIS (the one-wall reading); each face proved separately; the identification is the open core

Table 1 — The four laws of bounded observation: statement, corpus grounding, status

4 + 6 laws and theorems, every one stated and sourced	9 falsifiable predictions, each with an experiment	5 instrument-grade capabilities with protocols
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The laws are stated in Table 1 and the figure draws the ladder; the formal carriers follow in the next six chapters, one theorem per law where a theorem exists. Law 0 is the newest and the most falsifiable: it says that an observer at thermodynamic equilibrium — reversible belief dynamics, zero entropy production — cannot detect its own bottleneck from inside, so obstruction is strictly a nonequilibrium phenomenon. Law 1 is the conservation law: the corpus's no-go theorems were already its instances, and what is new is the invariant (the obstruction datum under transport) together with the cascade rigidity, the leakage multiplying as $1 + \delta(EF) = (1 + \delta(E))(1 + \delta(F))$. Law 2 is the production law and the arrow of time — DeepSeek's Continuation B, realized as a four-part theorem with one conjectural link. Law 3 converts the two walls of Volume I into an unattainability statement: the analytic wall (spectral tail) prices

the approach to zero obstruction, the combinatorial wall (compactness failure) prices the certificate search, and the one-wall hypothesis says they are one phenomenon.

Two reading rules keep the legislation honest, and both are enforced by the tables. First, no law is true by decree: each cites its grounding column, and the grounding column contains only proved corpus results or standard mathematics — where a law rests on a conjecture, the conjecture is named as such in the status column and attacked in the honesty register. Second, every law carries a predictive face: Law 0 is refuted by any equilibrium observer that detects its own bottleneck; Law 1 by any exact, conservative transport that destroys obstruction with zero leakage; Law 2 by any family in which obstruction persists at zero entropy production; Law 3 by any finite-price route to exact realizability past the spectral tail. Table 2 then maps the laws onto their formal carriers — the six new theorems — so that the reader can move from legislation to mathematics in one step, and from mathematics to the falsification table in two.

Theorem	Proved halves (corpus and here)	New content	Open core
I — Obstruction transport	Spectrum transport with tolerance overhead; isomorphism reflection of the five diagnostics; the intercept classification (no adjoint for nontrivial E); defect rigidity $\delta(e) = e^2 - 1$.	The conservation invariant; the reflection half; the multiplicative cascade law $1 + \delta(EF) = (1 + \delta(E))(1 + \delta(F))$ assembled as one statement.	A functorial statement in the cost-enriched graded category (blocked on the enrichment)
II — Graded small-gain	The Banach contraction 0.697 of the KM-averaged seven-map update; the per-optic Lipschitz bounds (0.92 six-fold, 1.15 expansion); the AAK floor side of the sandwich.	The input-to-state bound $D^* \leq c \cdot \Pi L / (1 - \Pi L)$; the relaxation rate $-\ln(\Pi L) \approx 0.361$ per cycle; the on-manifold/off-manifold pairing with the 3/2 fatigue law.	The equality $D^* = \sigma^*$ beyond the isometry case (Volume II's graded small-gain equality)
III — Distortion production	Cesaro existence (Birkhoff); the entropy chain-rule identities; the Schmidt–Mirsky tail floor; the Cherny–Spohn reversibility criterion.	The four-part theorem; the AAK-tail floor on the Cesaro rate; the arrow-of-time conjecture with five witnesses; DeepSeek's Continuation B realized.	Part (d3): obstruction implies entropy production, in general
IV — Non-analytic regulation	The metabolic side, fully proved: value–flux coupling, the $O(\epsilon)/O(\epsilon^2)$ dichotomy, Monge–Ampère atoms, 93.4–100% concentration, 66% non-reversion.	The quantum side: record phase transitions at replica-eigenvalue crossings; response concentration scaling inversely to the gap; the artefact discriminator for boundary authentication.	Finite-L smoothing of the predicted kinks (the L-ladder is the check)
V — Obstruction metrology	Affine dimension $d_A^2(nd_B^2 - 1)$; exact in-radius $2/(nd_B \min\{d_A, d_B\})$; the complete flat-width classification; the qutrit 4/3 exactly.	The obstruction index; the cascade calibration $r(n) = \lfloor (nd_B - 3)/2 \rfloor$; the spectrometer protocol; the 5/6 operational readout.	Hardware validation (Predictions 4 and 5)
VI — Optic interlacing	Zero-slice multiplicativity; rank and determination index multiply under independent blocks; the product-spectrum law for the unrestricted tensor lift.	Theorem 7.11's hypothesis discharged for independent blocks; Open Problem 7.13 split into a dichotomy; optic-Nehari isolated as the correlated-case lemma.	Optic-Nehari for correlated composites — the intrinsic structured case

Table 2 — The six new theorems: proved halves, new content, open core

3 · Theorem I — The Conservation of Obstruction

THEOREM I (Obstruction transport and conservation). Let P be a well-formed presentation at tolerance ε with obstruction datum $(\lambda_\varepsilon, \Gamma_{\text{set}} \supseteq \Gamma_{\text{eff}} \supseteq \Gamma_B)$, and let $F : P \rightarrow Q$ be a morphism of presentations carrying tolerance overhead ρ , an effective translator, and resource overhead H . Then three statements hold. (i) **PROVED-CORPUS**: obstructions transport forward — if ε lies in the obstruction spectrum of P at any of the outcomes L, D, K , then $\rho(\varepsilon)$ lies in the corresponding spectrum of Q , while successes transport with overhead; this is the spectrum-transport proposition of the monograph, the corpus’s own one-way valve. (ii) **NEW**: reflection prices relocation — if Q is unobstructed at $\rho(\varepsilon)$ within the transported budget while P is obstructed at ε , then F is not an isomorphism of obstruction data, and the tolerance gap $\rho(\varepsilon) - \varepsilon$ is exactly the ignorance F relocates; it vanishes if and only if F preserves and reflects all five diagnostic outcomes, which the monograph itself proves need not happen even when capability is preserved. (iii) **NEW**: the quantum instances are the sharp case — for an environment E of dimension $e > 1$, tensoring with E admits neither adjoint, the obstruction integer $e^4 - e^2 + 1$ lies strictly between $(e^2 - 1)^2$ and e^4 , and the minimal leakage is the adjunction defect $\delta(e) = e^2 - 1$, the dimension of $\text{su}(e)$, obeying the multiplicative cascade law $1 + \delta(E \otimes F) = (1 + \delta(E))(1 + \delta(F))$.

The proof posture is deliberately plain, because the theorem’s content is the invariant, not the machinery. Part (i) is proved in Effective Causal Descent v47 as the overhead spectrum-transport proposition; part (ii) is the contrapositive bookkeeping of the monograph’s transport and strictness propositions — capability equivalence does not imply diagnostic equivalence — assembled into a quantitative gap statement; part (iii) is the intercept classification of the combs and currying papers, joined with the adjunction-defect rigidity identity, stated as one law. Nothing is hidden. What is genuinely new is that three historically unrelated no-go families — the categorical intercept, the information-theoretic no-programming theorem, and the geometric Borsuk–Ulam compression floor $2/e^2$ — become three measurements of one conserved quantity. That is the move that turned “no perpetual motion machine works” into conservation of energy, and it is the move this theorem makes for bounded observers.

The physical reading is the law itself. Ignorance can be moved: sensor repair moves it into the controller — the monograph’s repair coordinates price exactly this, with zero-bit repair possible only for local behaviours, message alphabets of at least two, and detector-efficiency thresholds certified by linear programs with Farkas duals. It can be refined: the refinement–resolution bridge moves it into finer spatial resolution, and the rate–distortion formula prices the refinement. It can be priced: the obstruction’s exchange rate against states is the determination index. What no exact, conservative transformation can do is destroy it — and the non-exact ones leak measurably. The leak is not metaphor: under environment decoration, the hom-dimension of the one-slot comb jumps from 204 to 3132 on the two-qubit instance, the decoration rescales the source-linear coefficient while the constant term is rigid, and the deficit $e^2 - 1$ is precisely the non-trace-preserving directions that normalization removed. The first law of bounded

observation is thus also a bookkeeping law: every transformation of a constrained system files its ignorance somewhere, and the filing is auditable.

4 · Theorem II — The Graded Small-Gain Law

THEOREM II (Graded small-gain). Fix a resolution descent along a chain of n optics with per-optic Lipschitz bounds $L_1, \dots, L_n$, per-cycle distortion injection c , and AAK floor σ^* at the bottleneck width. Then: (i) **PROVED-CORPUS**: if the product $\prod L_i < 1$, the Krasnoselskii–Mann-regularized update is a Banach contraction with a unique fixed point — the unification object realized — uniformly in dimension; (ii) **PROVED-HERE**: the distortion accumulated at the fixed point obeys the input-to-state stability bound $D^* \leq c \cdot \Pi L / (1 - \Pi L)$, the geometric cascade of a contraction with injection, by summing the geometric series of attenuated injections; (iii) **PROVED-CORPUS**: $\sigma^* \leq D^*$, the AAK floor — no machine of the given width beats the operator-side tail, which is the lower half of Volume II’s sandwich; (iv) **NEW**: the approach to the fixed point is geometric with rate $-\ln(\Pi L)$ per cycle, and the excess $D^* - \sigma^*$ vanishes only in the distortion-neutral case — every optic an isometry on the bottleneck subspace with injection $c = \sigma^*$ — so the cascade’s price is positive in every other configuration and bounded by the same geometric series.

The corpus’s own instance saturates the theorem with unusual completeness, which is why the theorem is more than packaging. The seven-optic composition of the viability companion has per-optic bounds 0.92 (six contracting optics: $\alpha = 0.40$, $s = 1.00$, $\rho = 0.80$) and 1.15 (the RPSI expansion optic), so $\Pi L = 0.92^6 \cdot 1.15 = 0.697 < 1$, verified dimension-independently and numerically at worst case 0.674. The theorem then reads off two numbers the corpus does not state: the relaxation rate $-\ln 0.697 \approx 0.361$ per cycle, and the transient envelope $c \cdot 0.697^k$. **PREDICTION 1**: in the *E. coli* viability panels — where the corpus already measures cycle-scale behaviour across the iJO1366 and Keio data — the approach of the response to its asymptote across cycle size k should follow $1 - 0.697^k$, a semi-log slope of 0.361 per cycle. No such statistic exists in the manuscripts; the panels that would test it do.

The theorem also yields a new reading of a proved corpus result, and the reading is the pairing that makes the law feel physical. The companion proves the fatigue exponent: perturbing a loop of amplitude a by diffusion with variance rate $\sigma^2 = \nu a$ gives a correction $C_{\text{fat}} a^{3/2}$ with $C_{\text{fat}} = \sqrt{(5\nu)/2}$, measured at exponent 1.479 with $R^2 = 0.9999$ — the off-manifold response. The small-gain law gives the on-manifold response: geometric relaxation at 0.361 per cycle. The pairing says that a viability loop has exactly two response regimes — deterministic return along the contracting manifold, and diffusive fatigue transverse to it — with the exponent $3/2 = 1 + 1/2$ decomposing into the geometric factor 1 and the diffusive $1/2$. Neither manuscript connects these; the unification does, and the connection is falsifiable separately (Prediction 3: the exponent should be substrate-stable). Honesty: part (iv)’s equality analysis is proved only in the isometry case; the general equality $D^* = \sigma^*$ remains exactly the graded small-gain equality that Volume II isolated, and nothing here claims it.

5 · Theorem III — Distortion Production and the Arrow of Time

THEOREM III (Distortion production). Let X be a stationary ergodic finite-state process observed through a stationary channel; the observer maintains the Bayesian belief process b_t and acts through a resolution-limited summary of complexity M . Then: (a) **PROVED-HERE** — the Cesaro distortion rate $D_C = \lim (1/T) \sum_t d(b_t, \hat{b}_t)$ exists almost surely and equals the stationary expectation of the per-step distortion, by Birkhoff's theorem applied to the ergodic belief chain; (b) **PROVED-HERE** — for log-loss distortion, D_C equals the conditional entropy rate of the source given the resolution-limited observation, the gap to full observation is exactly the information the resolution drops, and $D_C = 0$ if and only if the summary is sufficient — which, on the linear relaxation, is precisely the corpus's zero threshold: $\text{rank}(H) \leq M$; (c) **PROVED-HERE** in the operator-norm form — the floor of any rank- M summary is the Schmidt–Mirsky tail σ_{M+1} , and in the scalar-alphabet Hankel-structured case the corpus's Theorem 7.9 upgrades the floor to the exact AAK identity $\Delta = \sigma_{M+1}$, with the multiletter case governed by Theorem 7.11's conditional and Theorem VI below; (d) the arrow of time: (d1) **PROVED-HERE** — the belief chain's entropy production, the Cherny–Spohn quantity $\sigma = \sum_i \pi_i k_{ij} \log(k_{ij}/k_{ji})$, is nonnegative and vanishes exactly when the belief chain is reversible; (d2) **PROVED-HERE** — a sufficient channel observing a reversible source gives $\sigma = 0$; (d3) **CONJECTURE**, the Resolution Second Law — whenever the obstruction is nonzero, $\sigma_{M+1} > 0$, the belief process is irreversible: $\sigma > 0$, and D_C is bounded by an explicit function of σ that vanishes at zero.

This is DeepSeek's Continuation B, realized to the exact depth the corpus supports, and the witnesses are what elevate (d3) from speculation to research programme. First, the metabolic pair: regulatory loops around closed parameter cycles scale linearly in loop size — measured slope 1.00 where smooth systems give 2.000 — and 66 percent of cycles do not revert: an empirical arrow of time at constraint-switch boundaries, priced by the value–flux coupling identity. Second, the viability companion: the $3/2$ fatigue law is a dissipation-like scaling for adaptation under repetition. Third, the quantum corpus: the replica interpolation identity $g(r) - r \cdot E \log X = \int_0^r (r-s) \text{Var}_s(\log X) ds$ says that the annealed–quenched gap is the monotone accumulation of variance along the interpolation — a one-way integral with no reversing operation, whose measured value is 0.0224, 0.0226, 0.0227 nats per site at $L = 8, 12, 16$. Fourth, the sustainability corpus: Cesaro aggregation conceals composition — the ECOMOD result that aggregate biocapacity can rise while ecological capital falls, in a bounded, demand-driven window ending in a floor collision. Fifth, the Born-record rate function of the monitored circuits: an entropy-production-shaped object pricing what an observer can infer from a measurement record, with the quenched large-deviation principle now proved in the corpus.

The Spohn connection, which DeepSeek named and Volume II only located, is now concrete: the Cesaro aggregation that the sustainability ledgers practice and the entropy production that nonequilibrium

physics defines meet in one quantity, the belief chain's σ , and the meeting has a sharp edge. Aggregation without entropy accounting is safe exactly at equilibrium — that is Law 0 — and it loses information at rate D_C everywhere else, which is the formal content of the ECOMOD warning: a scalar projection of a vector of components cannot be inverted, and the projection's information loss is precisely the distortion a bounded observer pays. The conjecture is falsifiable in simulation before it is attackable in proof (Prediction 6): hidden-Markov families with controlled Hankel tails, exact computation of both the tail and the belief-chain entropy production, and a search for any family with obstruction and $\sigma = 0$, or sufficiency and $\sigma > 0$. A single such family kills (d3); none found, and the second law of bounded observation stands with its witnesses running in a bacterium, a monitored circuit, and a compendium of planetary ledgers.

6 · Theorem IV — Non-Analytic Regulation and Record Phase Transitions

THEOREM IV (Non-analytic regulation). In a constrained-flow system, the second-order response concentrates on the non-analytic set of the value landscape, and its local magnitude is governed by the curvature concentrated there. The metabolic half is PROVED-CORPUS, and the corpus proves it with unusual completeness: the value-flux coupling identity $D^2\Phi = \sum_r c_r D^2v_r^*$ with the visibility dichotomy — eleven of twelve flux events objective-invisible on the reference cut, slope jumps up to 1.6×10^4 against value-layer movements below 1.5×10^{-7} ; the scaling dichotomy — curvature mass $B(\epsilon) = O(\epsilon)$ for piecewise-affine maps against $O(\epsilon^2)$ for smooth ones, measured 1.00 against 2.000 on the 76-pair scan; the Monge–Ampère atoms as determinants rather than products, the $2/\sin \theta$ overestimation law verified to 2.5×10^{-12} ; the refinement–resolution bridge carrying atoms to weak limits; and the headline empirics — 93.4 to 100.0 percent of total second-order response mass at active-set transitions across eleven crossing sweeps, ten of them exactly 1.000000, with a negative control at zero. The law's concentration set even has its geography: fifty reactions, 1.8 percent of the network, carry 85.9 percent of the curvature mass.

The quantum half is NEW, and it is this volume's headline claim of new physics: record phase transitions in monitored circuits. The disorder-averaged scaled cumulant generating function $\Lambda_L(\beta)$ of the Born record is computed, at $\beta = 1$ and $\beta = 2$, by the two- and three-replica compressed transfer operators, whose leading eigenvalues are real, nonnegative, and simple for all $0 < p < 1$. The corpus establishes an annealed critical chain — $p_c^{(2)} = 0.233810$, $p_c^{(3)} = 0.305(3)$, $p_c^{(4)} \approx 0.383$, $p_c^{(5)} \approx 0.47$ — a ladder of non-analytic points of the n -replica free energies in the measurement rate p , with the quenched critical point at 0.1597(8) sitting below; the production cells at $p = 0.16$ run immediately above it. The new statement: at each annealed boundary, the dominant eigenvalue families of successive replica operators exchange roles, and by standard eigenvalue perturbation theory the second-order response of the record statistics concentrates at the gap closures with magnitude scaling inversely to the gap — precisely the

metabolic signature, on a quantum substrate. The boundaries are certifiable because the closure is exact: the two-qubit Clifford group is an exact conjugation three-design to 1.6×10^{-14} , the two-replica anchor is reproduced to 10^{-14} , and the three-replica values $\Lambda(2) = -0.121924, -0.123857, -0.124526$ at $L = 8, 12, 16$ are identities, not estimates. PREDICTION 2: scanning p across the chain, the exact ESS exponent $\Lambda(2) - 2\Lambda(1)$ — today 0.0340, 0.0349, 0.0353 at $p = 0.16$ — develops derivative kinks at the certified boundaries, and the record-entropy density $\bar{x}_L$, today 0.1448 to 0.1489 bits per site, changes phase character across them.

The theorem's third face is a diagnostic the governance corpus needed but did not have: boundary authentication. The delay-dynamics and sampled-governance papers distinguish genuine instability crossings — the interval-certified Hopf window near 3.7 and 150 years, the Neimark–Sacker crossing near 6.5 years — from an artefact: an apparent 2.3-year threshold that dissolves under exact analysis as a discretisation artefact. The non-analytic regulation law supplies the general discriminator, because the law's signature is not merely a location but a magnitude: genuine crossings carry concentrated second-order response scaling inversely with the spectral gap, artefact crossings do not, and ECOMOD supplies the sustainability-side calibration — a constant leading eigenvalue (-0.0537 per year) along the drawdown branch certifies no critical slowing down, hence a floor collision rather than a fold. PREDICTION 8 makes the discriminator falsifiable: run the concentration test on both classes of crossings in the delay corpus and the *E. coli* sweeps; the law is refuted if it fails to separate them. A regulation law that also tells artefacts from realities is a technological capability, and it is counted as one in Chapter 10.

7 · Theorem V — Obstruction Metrology

THEOREM V (Obstruction metrology). The corpus's exact affine geometry of supplemented instruments is a measurement apparatus for the intercept, and it has never been read as one. Define the obstruction index ι_r of an instrument family at latent grade r as its flat-width ladder value δ_r — normalized so that $\iota = 1$ is flat. Then: (i) **PROVED-CORPUS** — $\iota_r = 1$ exactly in the flat classification ($nd_B \leq 4$ with $d_B \leq 2$ at grade 1; $n \leq 2r+2$ in the classical case; the qubit ladder), and $\iota_r \geq 4/3$ whenever $nd_B \geq 2r+3$, with equality exactly at $n = 2r+3$, at the qutrit ($d_B = 3, n \in \{1, 2\}$: $\delta_1 = \delta_2 = 4/3$ exactly), and at the classical five-outcome boundary; the top grade of prime-power dimension is $2(1-1/d_B)$. (ii) **DERIVED** — the cascade calibration: an n -outcome family on d_B has affine dimension $d_A^2(nd_B^2-1)$, exact in-radius $2/(nd_B \min\{d_A, d_B\})$ around the uniform depolarizing instrument — the optimal centre — and $\lfloor (nd_B-3)/2 \rfloor$ certified-flat latent grades, so the meter's dynamic range is exact in n and the reading is a lookup. (iii) **DERIVED** — the operational readout: $\iota = 4/3$ corresponds to single-use discrimination success $5/6$, so the obstruction index reports in operational units, not geometric ones.

The instrument — the obstruction spectrometer — is a four-step protocol, and every step rests on a proved formula. First, tomograph the instrument family: the affine body of the supplemented instruments

is a polytope whose exact dimension and radius the corpus computes. Second, measure the flat-width ladder against the depolarizing centre: the widths at each latent grade r , which the classification theory already fixes. Third, read the intercept grade off the calibration: flatness at a grade certifies the trivial-supplement condition there — the same condition the intercept classification equates with the existence of the adjoint — and every non-flat grade certifies obstruction with magnitude at least $4/3$. Fourth, certify with the exact formulas, in the arithmetic the corpus already practices. Sample readings, all exact: the qutrit state space reads $4/3$ at both bottom grades; the ququart lies in $[4/3, 3/2]$; the classical six-outcome simplex lies in $[4/3, 3/2]$; the full-scale reading is the decoration jump — a two-qubit comb's hom-dimension moving from 204 to 3132 under environment tensoring, the leak made visible as a number.

What makes this new physics rather than new accounting is the change of status it confers on the intercept. Before the theorem, the intercept principle was a theorem about what does not exist: no right adjoint, no universal evaluator, no finite memory that stores every channel — negatives, provable, but unobservable in principle. After it, the obstruction has a magnitude, a grade structure, an exact calibration table, and an operational unit; it is an observable awaiting hardware. The prediction is sharp enough to be embarrassing if wrong (Predictions 4 and 5): a qutrit instrument family, tomographed, must return exactly $4/3$ at both bottom grades — not approximately, because the value is an exact theorem, so the experiment measures the tomography, and the tomography measures the theory. This is the pattern the whole volume advocates for the quantum corpus: its exact closures (three-design to 10^{-14} , in-radius to the formula, widths to the classification) are not just proofs, they are calibration charts, and calibration charts are what instruments are made of.

8 · Theorem VI — Optic Interlacing and the Multiletter Ladder

THEOREM VI (Optic interlacing). Consider the multiletter problem of the transduction corpus — Theorem 7.11's conditional equality and Open Problem 7.13's demand for an intrinsic characterization. Then: (i) **PROVED-HERE**, zero slice — under independent block lifting, the realization ladder multiplies: $\text{rank}(H_1 \otimes H_2) = \text{rank}(H_1) \cdot \text{rank}(H_2)$, and the determination index is multiplicative under the corresponding direct-product composition, by the Myhill–Nerode characterization of the product automaton; consequently the zero threshold of the composite at block budget $M_1 M_2$ is exactly the product of the per-block thresholds, and the corpus's zero-slice law ($\text{rank}(H) \leq M$) transports intact. (ii) **PROVED-HERE**, positive slice — the singular values of the tensor lift are exactly the pairwise products $\{\sigma_i(H_1)\sigma_j(H_2)\}$, so the unrestricted grounded gap of the composite at any budget is the corresponding order statistic of the product multiset: the multiletter ladder is the product ladder, and the per-letter AAK identity (Theorem 7.9, $\Delta = \sigma_{M+1}$) transfers through it to the composite. (iii) **DERIVED**, the discharge — Theorem 7.11's hypothesis, a structure-preserving embedding carrying an AAK-type theorem, is satisfied

for independent blocks by the tensor embedding itself, with the approximant class read as the tensor of the per-letter structured classes; the conditional equality therefore holds there. (iv) CONJECTURE, optic-Nehari — for correlated letters, where the optic composite is not a pure tensor, the best structured approximant of a composite is the optic composite of best approximants; this is precisely the missing lemma of Continuation A, and the intercept leakage of Theorem I is what blocks its naive extension, because the correlated composite's hom-dimension need not factor.

The status change this effects on the corpus's own open problem is the theorem's substance, and it is worth stating exactly, because the corpus is scrupulous about its own conditionals. Theorem 7.11 assumes a unitary embedding of the multiletter Hankel into a free-semigroup or vector-valued Hardy class that carries a finite-rank approximation theorem — a hypothesis the corpus itself flags as strong and not known to be checkable in general, remarking that only the Eckart–Young–Mirsky lower bound is unconditional. Parts (i)–(ii) discharge that hypothesis for the independent-block case: the tensor product is the embedding, and the product-spectrum law is the approximation theorem on it — both elementary, both exact, neither previously assembled in the corpus's language. Open Problem 7.13, one undifferentiated demand for an intrinsic characterization, becomes a dichotomy: multiplicativity where the letters are independent — now proved — and optic-Nehari where they are not — now isolated as a named lemma with a precise statement. Reducing an open problem to a named missing lemma is the standard shape of progress in this corpus, and it is the shape delivered here.

The connection to Volume II is the reason this theorem carries the continuation's name. Volume II stated Continuation A as a proof strategy: block-length lifting is tensoring, tensoring is optic composition, so the multiletter ladder is the graded category's tensor ladder in disguise — and conjectured that the multiletter rung and the graded-trace small-gain law are one missing law in two dialects. Theorem VI makes the first half of that identification literal: the tensor ladder is now proved to carry the multiletter zero slice and positive slice for independent blocks, so whatever law closes the correlated case will be a statement about optic composites that are not tensors — exactly the class where the enrichment, the intercept, and the small-gain equality all live. Honesty compels two notes. First, the discharge in (iii) is partly definitional: choosing the structured class on the composite as the tensor of structured classes is a choice, and the intrinsic question — what the corpus's Remark 7.18 guards, that no Hardy-space embedding is assumed by default — remains open beneath it. Second, the independent-block case is the easy half, and it must not be advertised as the multiletter theorem; what it buys is a smaller, sharper target, and smaller sharper targets are how this programme has always moved.

9 · The New Physics

Four claims constitute the new physics, and the first is the headline. Record phase transitions (Theorem IV): the Born records of monitored quantum circuits carry a phase structure — a ladder of non-analytic points of the replica free energies, at $p_c^{(2)} = 0.233810$ and its higher-replica successors — at which the record statistics' second-order response concentrates with the same inverse-gap signature that metabolic regulation shows at active-set boundaries. The physics is new on two axes at once: no phase structure in measurement records of monitored circuits is established anywhere in the literature the corpus surveyed, and the quantitative identity of the two response laws — a classical LP substrate and a quantum replica substrate — is a unification claim that only the corpus's two halves together could even state. It is also certifiable physics: the boundaries are computed by an exact closure whose anchors reproduce to 10^{-14} , so the prediction can be checked without new theory, on production runs the corpus already maintains, and the v23 L = 8 rung currently computing is the natural first scan.

Second, obstruction as a physical observable (Theorem V): the intercept dimension, previously a negative theorem, becomes a measurable quantity with an exact calibration — the obstruction index, quantized by dimension combinatorics ($4/3$ at $n = 2r+3$, at the qutrit, at the classical boundary), with an operational readout in discrimination success and a hardware protocol attached. Third, the one-wall hypothesis (Law 3's core): the combinatorial wall and the analytic wall are two faces of one compactness phenomenon. The evidence is structural: the Eckart–Young–Mirsky and AAK floors are compactness statements about the approximant family; weak König's lemma is the failure of uniform compact search; the Borsuk–Ulam bound converts compactness into a diameter floor; the Weihrauch spectrum — with the halting-stage game's obstruction sitting exactly at the PA degrees, and recursive capability complete for Σ^3 — grades the search costs. The reading this suggests for the halting-physics corpus is the genuinely new sentence: the physics of halting is the thermodynamics of compact search, with the PA-degree spectrum playing the role of a temperature scale. Fourth, the curvature–regulation correspondence as cross-domain physics: one quantitative law — second-order concentration at the non-analytic set, magnitude inverse to the gap — now instantiated on metabolic networks, monitored circuits, and governance delay dynamics, with the artefact discriminator as its practical test.

4/3 the qutrit obstruction index — exact	0.233810 the first record-phase boundary — certifiable	0.361 the E. coli cascade relaxation rate — per cycle
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The discipline that keeps these four honest is the same one the corpus practices on itself, applied at the level of claims rather than proofs. Record phase transitions are a theorem in the two-level fragment and a prediction in general, and the prediction carries its own finite-size check — the L-ladder distinguishes genuine kinks from finite-L smoothing, exactly as the corpus's finite-size scaling separated the quenched

critical point 0.1597(8) from the annealed chain. The observable-obstruction claim is derived, but its hardware test is a genuine experiment, and its failure mode is informative either way: a qutrit reading off 4/3 confirms the tomography and the theory together; anything else localizes the defect in one of them. The one-wall hypothesis is the only claim of the four with no proof face at all — it is the volume’s one piece of pure programme physics, the compactness identification stated so that someone can refute it by exhibiting a compactness modulus that does not control a spectral tail, or prove it by exchanging one for the other. It is listed as such, and it closes this chapter the way the third law closes the ladder: unreachable, priced, and exact about both.

10 · The New Capabilities

Capability one: certified large-deviation telemetry for monitored quantum circuits. The exact closure turns the ESS law — $\text{ESS}/B = \exp[-2Lt \cdot (\Lambda(2) - 2\Lambda(1))]$ — from a warning into an engineering budget curve: at $p = 0.16$ the exponent is 0.0340 nats per site at $L = 8$, and the true ESS fractions are 2.7×10^{-8} , 3.6×10^{-18} , 4.2×10^{-32} at $L = 8, 12, 16$, so any fixed sampling budget fails at a computable system size, and the computation is exact. The self-averaging criterion — the per-site variance of the record count, 0.104–0.106, L -independent — is the sampling certificate that says when a single disorder realization speaks for the ensemble; the record-entropy density ladder (0.1448 to 0.1489 bits per site) is the telemetry channel itself. The protocol: run the monitor, report the record exponents with exact-arithmetic certificates attached, flag the record-phase boundaries by the certified p -locations. This is the first capability in the quantum corpus that an experimentalist could operate without reading the theory, because every number it reports is either exact or carries its own error contract.

Capability two: the obstruction spectrometer (Theorem V’s protocol, with its calibration table). Capability three: curvature-guided intervention design. The design rule falls directly out of Theorem IV’s proved metabolic half: interventions placed at curvature atoms — the non-analytic boundaries where second-order response concentrates — buy maximal response per unit cost, because the value–flux coupling identity fixes the exact exchange rate there, and the epistasis structure ($\rho_S = 0.865$ between curvature epistasis and footprint overlap) supplies the interaction bookkeeping. The sustainability corpus supplies the calibrated instances: the COD intervention analysis returns a largest robust constant catch of 91.6 kt and a certified-kernel horizon of seven years at the limit reference point; the Edwards analysis returns a smallest securing cut of 7.2 percent against a 618-ft threshold; and — the negative instance that proves the rule — the reactive rules that are “invisible to the kernels of their own triggers” are interventions placed off the concentration set, failing silently exactly where the curvature is not. The frozen-protocol forecast ladders are the falsification harness, and Prediction 9 states the dominance test.

Capability four: dual-certified policy bands — machine-checkable governance. The minimax dual certificate theorem says the common safe-action set is empty precisely when an adversarial probability measure on the boundary–disturbance bundle drives expected inward drift negative against every control,

with support of at most $k+1$ points, tight; the computational certification bridge turns it into a finite linear program whose optimum brackets the continuous-time safety value from both sides with certified error inflation. Assembled, these are policy proofs: a policy certificate is a small rational object — a measure with at most $k+1$ atoms, an exact margin (the benchmark instance certifies obstruction at $Y = 6$ with margin $3/50$ on the two-stock problem whose critical aggregate is exactly $27/5$) — that any third party can verify without re-solving anything, in the exact arithmetic the corpus already practices. This is the sustainability corpus's answer to its own reception risk: not more models, but certificates that travel. Capability five, stated briefly because the chess paper already carries it: role-decomposed theory engines. The CSR sufficiency theorem and the amplification bound $P \geq [\theta(1-(1-q)^w)]^L$ are substrate-independent; the corpus's own arena agents are running instances; and the society architecture — generator, critics, verifier, attacker, coordinator — is the operational form of this volume's own discipline of separating the proved from the conjectured.

11 · The Falsification Table

A claim to new science is a contract with its refuters, and this volume signs nine. The table states each prediction with its experiment, its observable, its predicted value, and the verdict that would count — the same contract format the corpus's own claims-evidence maps practice, extended from claims to creations. Three of the nine run on data that already exists (the *E. coli* panels, the delay-dynamics models, the ECOMOD branch), four run on instruments the corpus already maintains (the monitored-circuit production runs, the intervention models, the belief-state exact-arithmetic engines), and two require new hardware or new simulation (the spectrometer readings, the arrow-of-time families). None of them requires new theory to check, which is deliberate: a prediction that cannot be tested without first proving something is not a prediction but a promissory note, and the corpus has always refused those.

#	Prediction	Experiment and observable	Verdict
1	The <i>E. coli</i> viability cascade relaxes geometrically: approach-to-asymptote follows $1 - 0.697^k$ over cycle size k .	Cycle-size scan on the iJO1366/Keio panels; semi-log slope of response vs cycle index.	Slope within confidence bounds of $-\ln 0.697 \approx 0.361$ per cycle
2	Record phase transitions: the exact ESS exponent $\Lambda(2)-2\Lambda(1)$ develops derivative kinks at the annealed boundaries.	Scan p across 0.16–0.5 in the monitored-circuit production runs (Clifford, $d = 2$); boundaries at $p_c^{(2)} = 0.233810$, $p_c^{(3)} = 0.305(3)$, $p_c^{(4)} \approx 0.383$.	Kink localization inside the certified windows, on the L-ladder
3	The off-manifold fatigue exponent $3/2$ with $C_{\text{fat}} = \sqrt{(5v)/2}$ is substrate-stable.	Repeat the perturbed-loop Monte Carlo on new networks (already 1.479, $R^2 = 0.9999$ on the reference).	Exponent within the bootstrap CI [1.47, 1.49] on new substrates
4	The qutrit obstruction index is exactly $4/3$ in hardware.	Instrument-family tomography; compute the flat-width ladder against the depolarizing center.	$\delta_1 = \delta_2 = 4/3$ within experimental error

#	Prediction	Experiment and observable	Verdict
5	Cascade calibration: n-outcome families show $L(nd_B - 3)/2J$ certified-flat latent grades.	Flat-width ladder measurement across n and d_B ; in-radius decay $2/(nd_B s)$.	Ladder matches the exact formulas
6	No counterexample to the arrow of time: filter entropy production is strictly positive whenever the AAK tail is nonzero.	Hidden-Markov simulation families with controlled Hankel tails; compute belief-chain EP exactly.	Any sufficiency-with-EP>0 or obstruction-with-EP=0 family refutes (d3)
7	Equilibrium blindness: at zero entropy production an observer cannot detect its own bottleneck from inside.	Simulation: reversible sources with planted obstructions; measure internal detection rate.	Detection at chance level confirms; above-chance detection refutes Law 0
8	Boundary authentication: genuine crossings carry concentrated second-order response; artefact crossings do not.	The delay-dynamics corpus: Hopf window 3.7–150 yr and Neimark–Sacker 6.5 yr against the artefact 2.3-yr threshold; E. coli sweeps as calibration.	Concentration separates real from artefact in every instance
9	Curvature-guided interventions dominate uniform interventions on response per unit cost.	COD and Edwards intervention models (robust catch 91.6 kt; smallest securing cut 7.2%) plus the E. coli epistasis panels ($\rho_S = 0.865$).	Dominance on the declared metrics; silent failure off the concentration set

Table 3 — The falsification table: nine predictions, each with an experiment and a verdict

The verdict conditions are asymmetric by design, because the claims are asymmetric. Predictions 2, 4, and 5 test exact values, so they measure the apparatus as much as the theory — a kink that fails to localize inside the certified window, or a qutrit ladder that misses $4/3$, falsifies cleanly. Predictions 1, 3, and 9 test scaling laws, so they carry confidence intervals rather than equalities — the 0.361 slope within bounds, the $3/2$ exponent inside its bootstrap interval on new substrates. Predictions 6, 7, and 8 test the conjectural and diagnostic faces, and they are refuted by single counterexamples or single classification failures: one family with obstruction and zero entropy production kills the Resolution Second Law; one equilibrium observer that detects its own bottleneck kills Law 0; one crossing class the discriminator cannot separate kills the authentication capability. The asymmetry is the honesty register made operational: the further a claim sits from proof, the easier the volume has made it to kill.

12 · The Provenance Ledger

The ledger closes the loop the mandate opened. Every new claim in this volume — four laws, six theorems, four physics claims, five capabilities — is listed against its corpus sources, cited by label and version, with the novelty delta stated in the third column as a countable difference: what was added, and of what kind. The ledger is the volume’s answer to the question DeepSeek put to the entire programme, and that Volume II put to Volume I: does the unified framework prove something new? The answer given here is not a yes but an inventory — new statements, discharged hypotheses, assembled invariants, unmeasured numbers, unbuilt instruments — because an inventory can be audited and a yes cannot. An auditor with access to the twelve repositories can verify every row: the sources exist, the labels resolve, the deltas are differences between what the rows say and what the sources say.

New claim	Corpus sources (label, version)	Novelty delta
Law 0 — equilibrium blindness	no-freeze degenerate case (qc v22); ECOMOD eigenvalue (v37); profinite duality (rd v9, Meta-Thm 4.1); Spohn/Cherny criterion (standard, via Continuation B)	the statement; the duality obstruction-reversibility; Prediction 7
Law 1 / Theorem I — conservation of obstruction	def:obstruction-datum, prop:obstruction-transport, prop:overhead-spectrum-transport (ECD v47); thm:chan-no-right, thm:instr-no-left, thm:exact-classification (currying/combs); thm:no-universal-evaluator; rem:adjunction-defect	the conserved-quantity framing; the reflection half; the multiplicative cascade law as one statement; the 204→3132 leak reading
Theorem II — graded small-gain	thm:unconditional-banach, lem:lip-per-optic (companion v15); AAK floors (rd v9); the sandwich (Vol II BT2)	the ISS bound; the 0.361 relaxation rate; Prediction 1; the on/off-manifold pairing with thm:levy-3half
Theorem III — distortion production	Thm 6.6, Prop 7.8, Thm 7.9 (rd v9); prop:replica-int (qc v22); thm:levy-3half, prop:dichotomy (companion v15, main v22); ECOMOD v37	the four-part theorem; the AAK-tail floor on the Cesaro rate; Continuation B discharged in the finite case; Prediction 6
Theorem IV — non-analytic regulation	thm:coupling, prop:dichotomy, prop:maatom, thm:Bprime, the 93.4–100% sweeps (main v22); thm:rank, prop:z3closure, prop:cliffscgf, the p_c chain (qc v22); paper4/paper5 (GS Round 32)	the quantum side; record phase transitions; Prediction 2; the artefact discriminator (Prediction 8)
Theorem V — obstruction metrology	prop:dimension, thm:exact-radius, cor:bottom-dichotomy, thm:state-nonflat, cor:flat-fails, cor:exact-d1, thm:mass-exact (instruments v18)	the obstruction index; the cascade calibration; the spectrometer protocol; Predictions 4 and 5
Theorem VI — optic interlacing	Thm 7.9, Thm 7.11, Open 7.13, Prop 7.8, Thm 6.6 (rd v9)	zero-slice multiplicativity; the product-spectrum law; the discharge of 7.11’s hypothesis for independent blocks; optic-Nehari isolated
Record phase transitions (physics claim)	the replica transfer family and annealed critical chain (qc v22); exact three-replica closure to 10^{-14}	the phase structure in Born records; the kink prediction; the certification route
Obstruction as observable (physics claim)	the exact affine-geometric classification (instruments v18); the intercept principle (combs/currying)	the meter: obstruction read off hardware for the first time

New claim	Corpus sources (label, version)	Novelty delta
One-wall hypothesis	WKL/PA/ Σ^3 spectra (ECD v47); AAK floors (rd v9); PostBQP (qc v22); Borsuk–Ulam $2/e^2$ (currying)	the compactness identification; the reading of halting physics as thermodynamics of compact search
Five capabilities	ESS law and Z_3 closure (qc v22); affine formulas (instruments v18); minimax dual certificates v11; LP two-sided bridge v18; CSR and Theorem 2* (chess)	the protocols: certified telemetry, spectrometer, curvature-guided intervention, policy proofs, role-decomposed engines

Table 4 — The provenance ledger: every new claim against its corpus sources and its novelty delta

Two patterns in the ledger are worth reading off, because they describe the shape of the volume’s contribution better than any summary sentence. First, no row has an empty source column: nothing here is free-floating — every creation is grounded in proved mathematics that was already sitting in the corpus, which is exactly what distinguishes this constructive turn from the juxtaposition risk the whole programme has been warned against since DeepSeek’s first ruling. Second, no row has an empty delta column either: nothing here is a restatement — every row adds a statement, a discharge, an invariant, a number, or a protocol that the source manuscripts do not contain. Grounded novelty, row by row, is the strongest form of “new” that honest synthesis can deliver, and the ledger is its proof of work.

13 · The Honesty Register

The strongest attacks on each new object, stated as an adversary would state them. Against Law 0: an equilibrium observer might detect its bottleneck through a channel the reversibility criterion does not cover — a driven-but-sufficient source with positive entropy production and zero obstruction would leave the law standing but gut its converse; the simulation search of Prediction 7 is where this is settled. Against Theorem I: the conservation form is assembled from proved transport propositions, but the functorial statement — the one that would make “conserved” precise in the graded category — is blocked on the enrichment that Volume II named and this volume does not build; until then, the first law is a family of instances with a common bookkeeping, not yet one theorem. Against Theorem II: the equality analysis is proved only in the isometry case, and the pairing of the 0.361 relaxation with the $3/2$ fatigue is a reading, not a derivation — the two laws govern the same loop in different regimes, but no lemma here derives one from the other.

Against Theorem III: parts (a)–(c) are standard mathematics applied to corpus objects, and their novelty is the assembly; part (d3) is a conjecture whose witnesses are correlational — the metabolic arrow, the fatigue exponent, the replica gap, the ECOMOD window, and the record rate function all point the same way, but none of them entails the implication, and a lumpable but driven counterexample family is a live possibility. Against Theorem IV: the kink prediction could be smoothed by finite-L effects rather than refuted by them, which is why the prediction includes the L-ladder as its own control; the metabolic half is proved, the quantum half is proved only at the level of eigenvalue-crossing perturbation in the

fragment, and the cross-substrate identity of the two response laws is an extrapolation with two data points. Against Theorem V: the meter presumes the affine model of the instrument family, and a hardware deviation measures the tomography before it measures the theory. Against Theorem VI: the independent-block discharge is the easy half, partly definitional, and must not be read as the multiletter theorem; against the one-wall hypothesis: it has no proof face, and it is stated so that its refutation is a well-defined task. None of the six theorems claims its open core; the cores are in Table 2.

What this volume does not claim, finally, because the programme's credibility rests on these denials holding. It does not claim BT1a — the obstruction-to-Hankel recovery remains the deepest open link, untouched here. It does not claim the cost-enriched graded category exists — it remains the load-bearing gap, and two of the six theorems (I and VI) would upgrade from assemblies to single functorial statements the day it is built. It does not claim any cohomological, homotopical, or higher-arity completeness for the obstruction datum — the monograph's own scope guard stands, and DeepSeek's Bridge Theorem 1 remains corrected, not vindicated. It does not claim the multiletter AAK theorem, the arrow-of-time theorem, or the graded small-gain equality — it claims discharged fragments, named lemmas, and falsifiable conjectures in their direction. And it does not claim that any prediction has been tested: the nine entries of Table 3 are offers, not results. The volume creates at the frontier; the frontier moves; the ledger records where it stopped.

14 · Research Orders

The priority queue is ordered by information per unit cost, and the first three orders are cheap because the apparatus already exists. Order one: run the record-phase scan. The monitored circuits are solved, the closure is exact, the boundaries are certified at 0.233810 and above, the production cells at $p = 0.16$ sit a step above the quenched critical point, and the $v_{23} L = 8$ rung is computing now — a p -scan across the annealed chain with the exact exponent ladder attached is the fastest falsification or confirmation of this volume's headline physics. Order two: the *E. coli* cycle-size scan — a new statistic (approach-to-asymptote against $1 - 0.697^k$) on panels that already exist. Order three: state and attack optic-Nehari for correlated composites — a named lemma with a precise statement, the one proof that would upgrade a published conditional theorem (7.11) and split a published open problem (7.13) permanently; it is the highest-value proof target in the queue because Volume II's Continuation A and the graded-trace small-gain law both converge on it.

Orders four through seven are the build-and-prove tier. Four: write the obstruction-spectrometer protocol as a short instrument paper — every formula is proved in the instruments corpus, the calibration table is exact, and the qutrit prediction is a clean experimental contract. Five: the arrow-of-time simulation campaign of Prediction 6, which either kills part (d3) cheaply or arms the theorem that Continuation B was pointing at. Six: the enrichment — unchanged in priority from Volume II, still the load-bearing gap, and now carrying two upgrade promises (Theorems I and VI) instead of one. Seven: BT1a, the deep

identification, last as ever. The staging respects DeepSeek's path with the same correction Volume II made: the bridge paper first, fifteen to twenty pages, but its skeleton now has meat that the earlier volumes could not give it — a proved fragment of Continuation A, a realized finite-case of Continuation B, a conservation law with instances across eight domains, and a physics prediction that a running computation can check this month.

If you can prove even one, you have a major contribution. If you can prove all three, you have a unified theory of viability, information, and composition.

That was DeepSeek's closing verdict on the programme, and this volume answers it in the only currency that verdict accepts. Six theorems are stated, each with its proved halves separated from its open core; two discharged fragments advance the two continuations it offered; four laws legislate the no-go theorems it wanted unified; nine predictions stand ready for their refuters; five instruments wait for their operators. The corpus's own epistemology — charts carry certificates, ladders carry monotonicity, limits carry invariants — predicts what these creations are: the invariants the limit was carrying all along. The climb that Volume II ended by recommending continues here with handholds it did not have: the ladder is the same, but three more rungs hold weight, and the next one is named.